A
Panel
WES
WGS
SnpEff
fastVEP
VEP
1.06×
2.66×
10.78×
0.46×
0.93×
2.54×
0.71×
2.29×
9.50×
5 s 15 s 1 min 5 min 30 min2.0 h
Median end-to-end runtime
B
Panel
WES
WGS
Speedup over direct annotation
20×
10×
5×
2×
1×
0.5×
0.25×
VEP
fastVEP
SnpEff
KPGP
SGDP
C
SnpEff
fastVEP
VEP
1.06×
6% saved
2.66×
62% saved
10.78×
91% saved
0.46×
118% longer
0.93×
7% longer
2.54×
61% saved
0.71×
42% longer
2.29×
56% saved
9.50×
89% saved
Panel
WES
WGS